

TSFS12 Hand-in 4: Collaborative Control

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Exercise 5.1 – Position-Based Formation Control

Double-integrator implementation

Each agent is modeled as a double integrator with the state

$$x = [p_x, p_y, v_x, v_y]^T.$$

The system dynamics are modeled as

$$\dot{p}_i = v_i, \quad \dot{v}_i = u_i + \frac{w}{m},$$

where u_i is the control input and w is the constant wind force. The controller uses proportional feedback as requested by the exercises with gains configured to $K_p = K_v = 4$. The default starting positions and trajectories were used.

Wind trajectory comparison

Without wind, the agents nicely approach their target trajectory after an initial transient from their starting positions, as shown in Figure 1. With a constant wind force the trajectories become displaced from their references, see Figure 2. At steady state, the position error balances the wind force. Since the model parameters were set to $w = [4, 0]^T$, $m = 1$, and $K_p = 4$, the displacement becomes

$$\Delta p = \frac{w}{mK_p} = [1, 0]^T,$$

so the wind shifts the agents one unit in the x -direction.

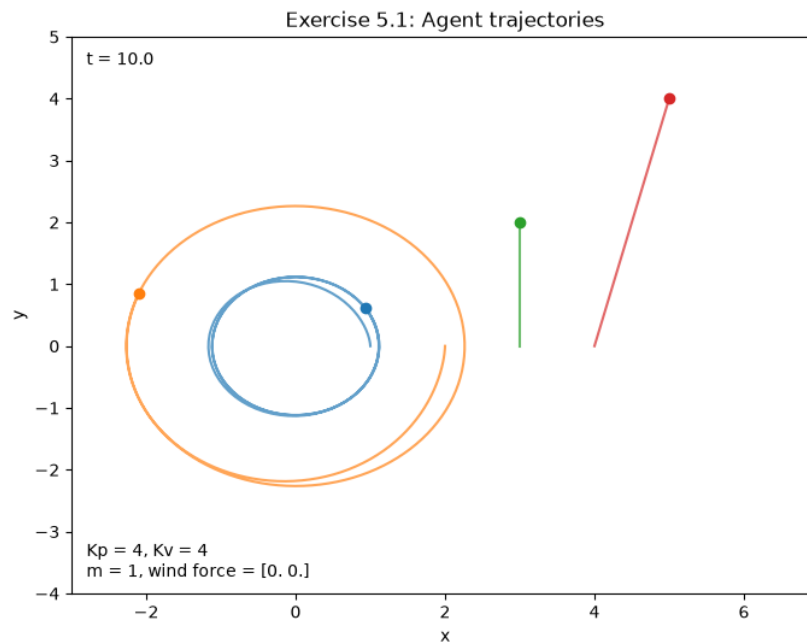


Figure 1: Agent trajectories without wind.

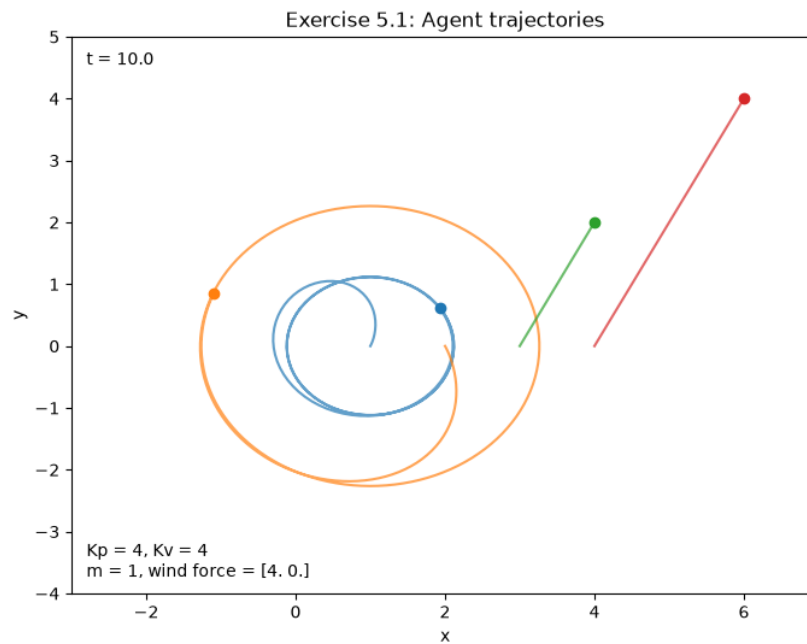


Figure 2: Agent trajectories with a constant wind force.

Exercise 5.2 – Displacement-Based Formation Control

Task 3

The simulations show that the agents form the desired vertical string for both $\omega = 1$ rad/s and $\omega = 2$ rad/s. Initially, a large transient occurs as the agents move from their starting positions to the formation. After the transient, the motion of the top agent propagates down through the directed chain.

Figures 3 and 4 show the agent trajectories for $\omega = 1$ rad/s and $\omega = 2$ rad/s respectively. The trajectories are quite similar in the two cases, since the position oscillation amplitude is small compared to the initial transient.

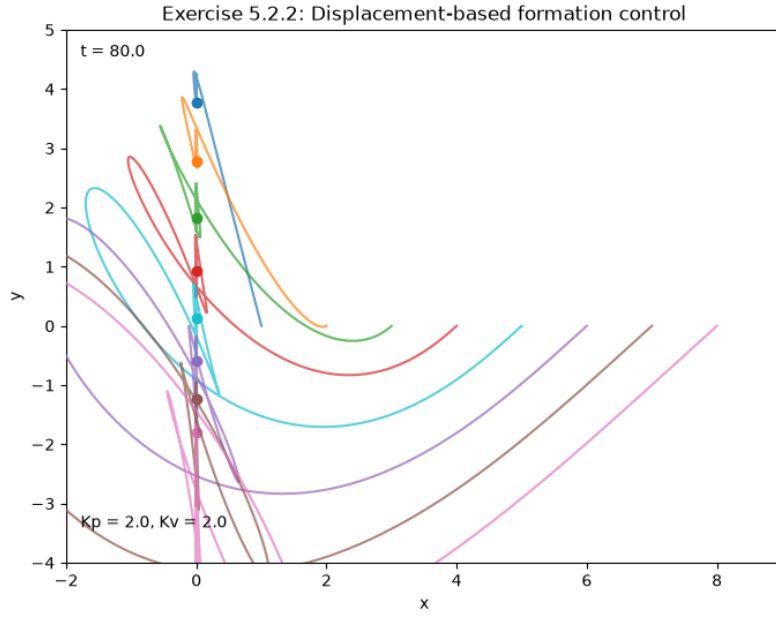


Figure 3: Agent trajectories for $\omega = 1$ rad/s.

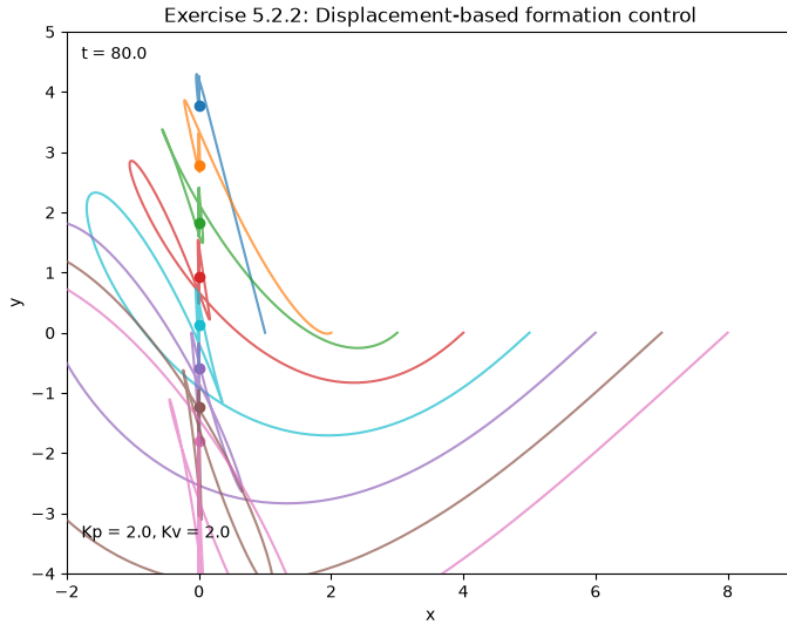


Figure 4: Agent trajectories for $\omega = 2$ rad/s.

The difference between the two cases is more clearly seen in the vertical velocity plots in Figures 5 and 6. For $\omega = 1$ rad/s, the velocities oscillate more slowly, while for $\omega = 2$ rad/s the oscillations are faster and have a larger velocity amplitude. This is consistent with

$$v_y^*(t) = 0.2\omega \cos(\omega t),$$

which gives velocity amplitudes of 0.2 m/s and 0.4 m/s for $\omega = 1$ and $\omega = 2$, respectively. Phase differences between the agents can also be seen, since the motion propagates through the directed chain from the top agent downwards.

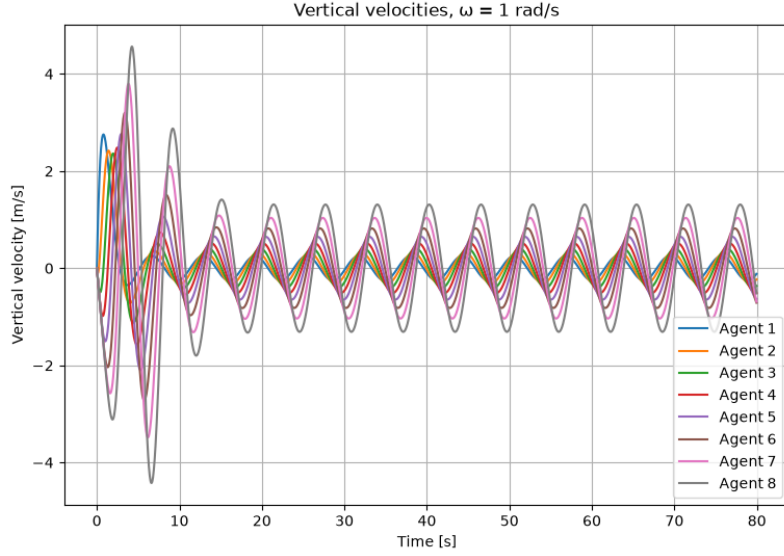


Figure 5: Vertical velocities of the agents for $\omega = 1$ rad/s.

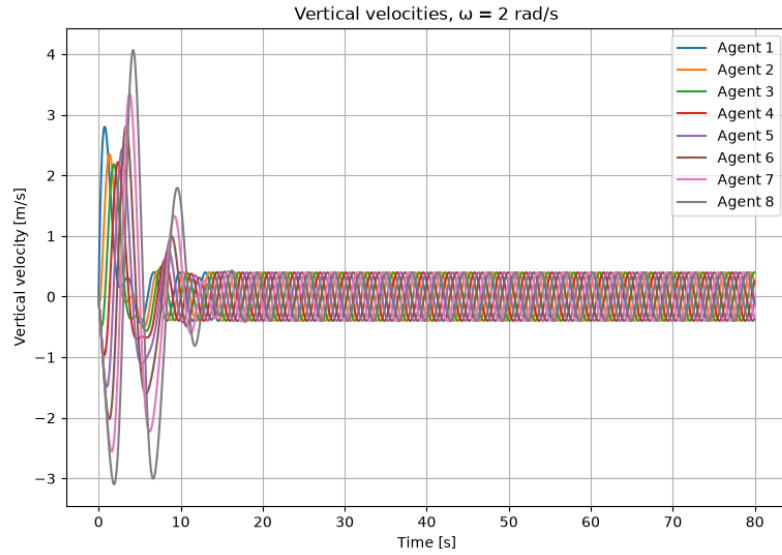


Figure 6: Vertical velocities of the agents for $\omega = 2$ rad/s.

Task 4

The signs of both control laws were changed and the simulation was repeated. With the original control laws, the agents converge towards the desired vertical formation, as shown in Figure 7.

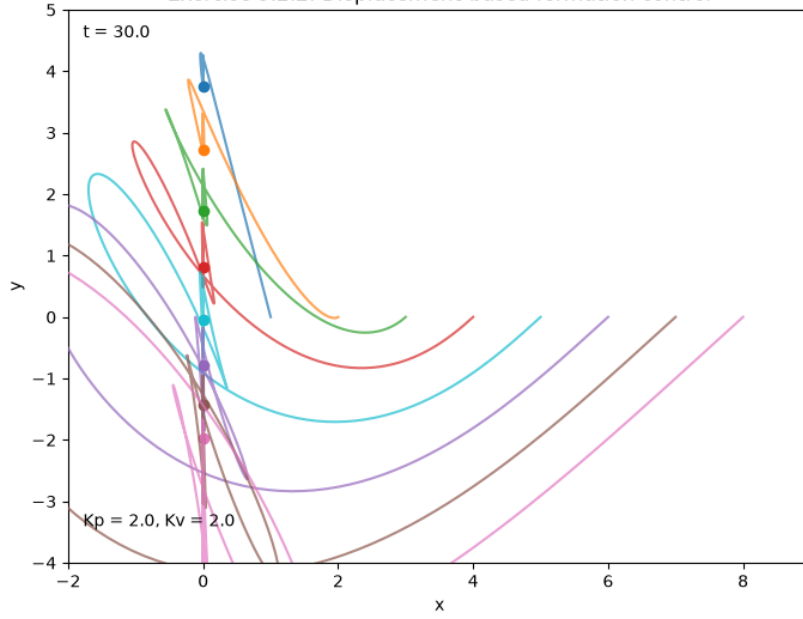


Figure 7: Agent trajectories with the original control-law signs.

When the signs of the control laws were reversed, the behavior changed significantly. Instead of converging to the desired formation, the positions of the agents increased rapidly and the system diverged. The position values reached very large magnitudes during the simulation, as shown in Figure 8.

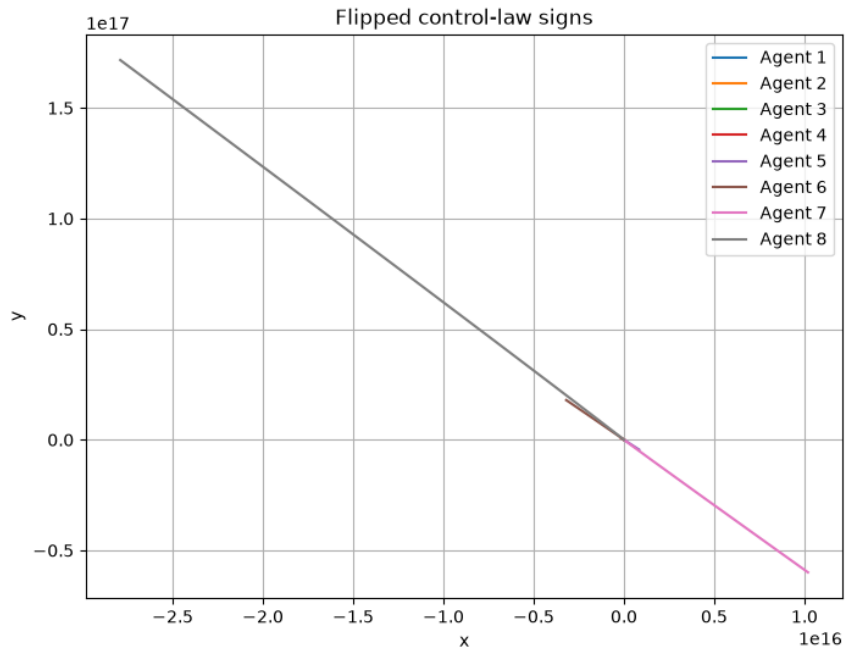


Figure 8: Agent trajectories after reversing the signs of the control laws.

From the simulations, it was therefore observed that the original signs give a converging formation, while reversing the signs causes the agents to move away from the desired formation and the system to diverge.

Exercise 5.3 – Distance-Based Formation Control

Task 1

A minimally rigid graph was used to specify the distance constraints for the agents. The top agent acts as the reference agent and as a precomputed circular trajectory of radius 2, while the remaining agents try to preserve their distance to its neighbors. The simulation was performed with $k_p = 1$ and $k_p = 5$. To compare the two cases, the distance between the top agent and its first neighbor was measured. The target distance for this pair is approximately $\sqrt{2} \approx 1.41$, marked with a red line in the plot.

Figure 9 shows the result for $k_p = 1$. The agents follow the moving formation, but the distance between the top agent and its neighbor oscillates quite significantly in both directions during the course of the motion.

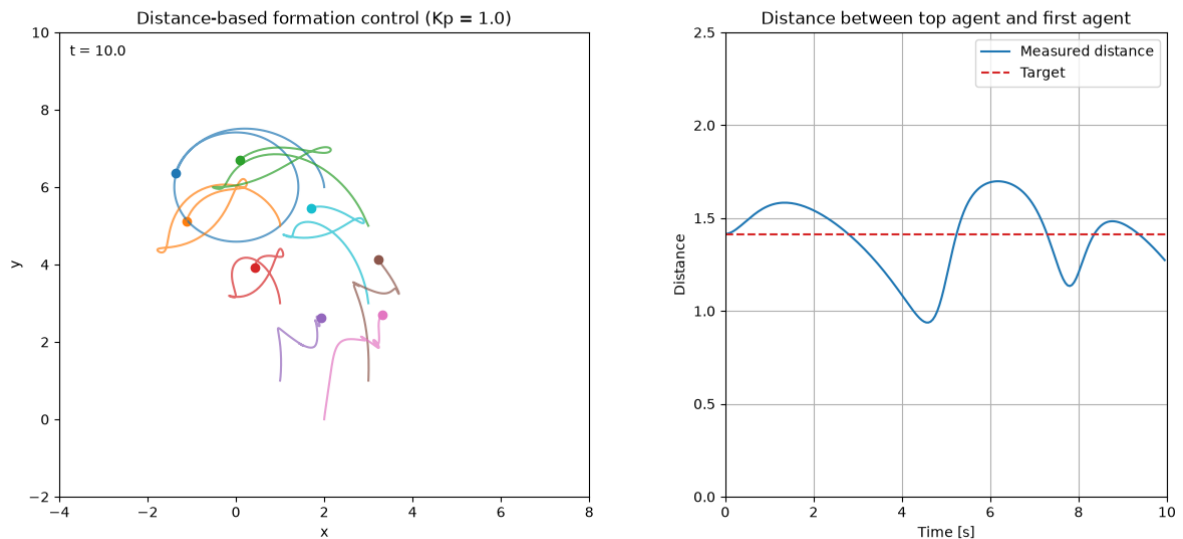


Figure 9: Distance-based formation control with a moving top agent for $k_p = 1$.

Figure 10 shows the corresponding result for $k_p = 5$. In this case, the measured distance stays much closer to the desired value. There are still some errors present, but they are clearly reduced compared to the previous case.

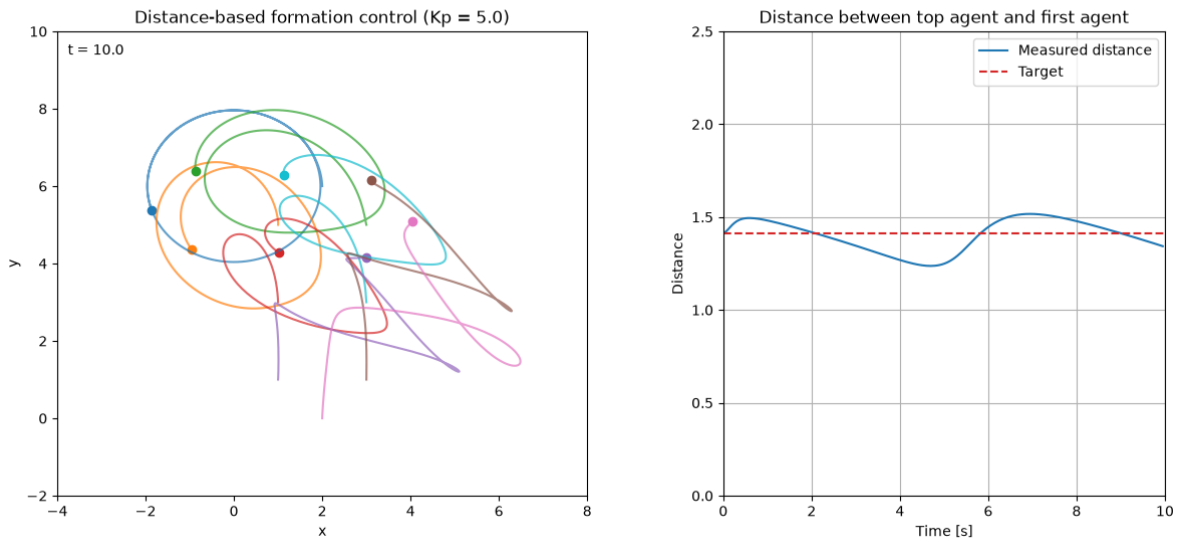


Figure 10: Distance-based formation control with a moving top agent for $k_p = 5$.

The main difference between the two simulations is how quickly the formation reacts to changes caused by the moving top agent. With $k_p = 5$, the agents react more strongly and are therefore able to maintain the desired formation distances more precisely.

Task 2

Making the bottom agent also use absolute positioning showed to enable the formation to keep its upright orientation.

Figure 11 shows the result for $k_p = 1$. The agents follow the circular motion while maintaining the overall orientation of the desired formation. However, some deformation can still be seen because the lower controller gain gives a slower response but is more damped compared to when only the top agent used absolute positioning.

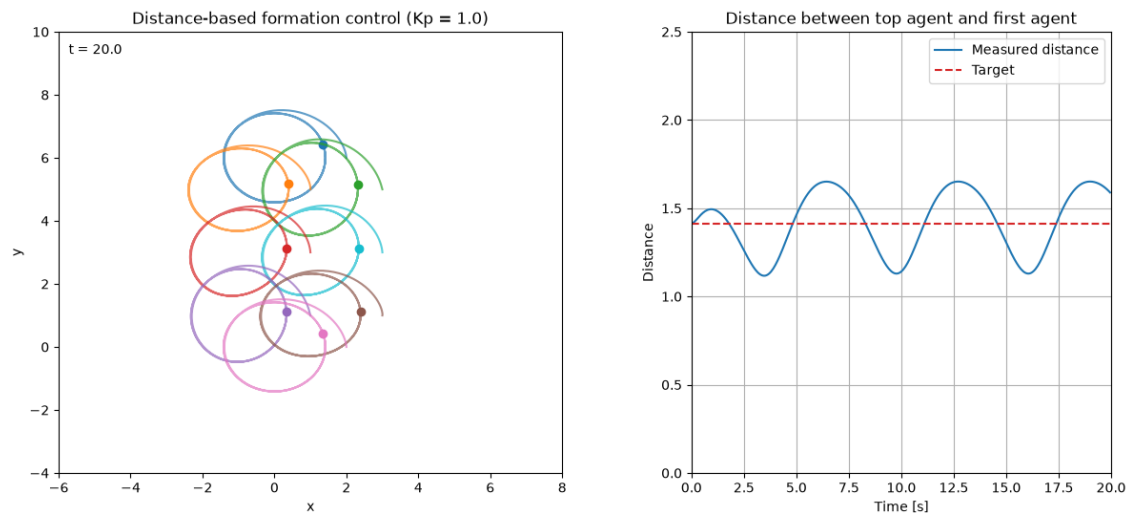


Figure 11: Agent paths with absolute position control of the top and bottom agents for $k_p = 1$.

For $k_p = 5$, shown in Figure 12, the formation follows the desired motion even more closely. The agents stay closer to their desired relative positions, and the orientation of the formation is maintained more accurately throughout the motion.

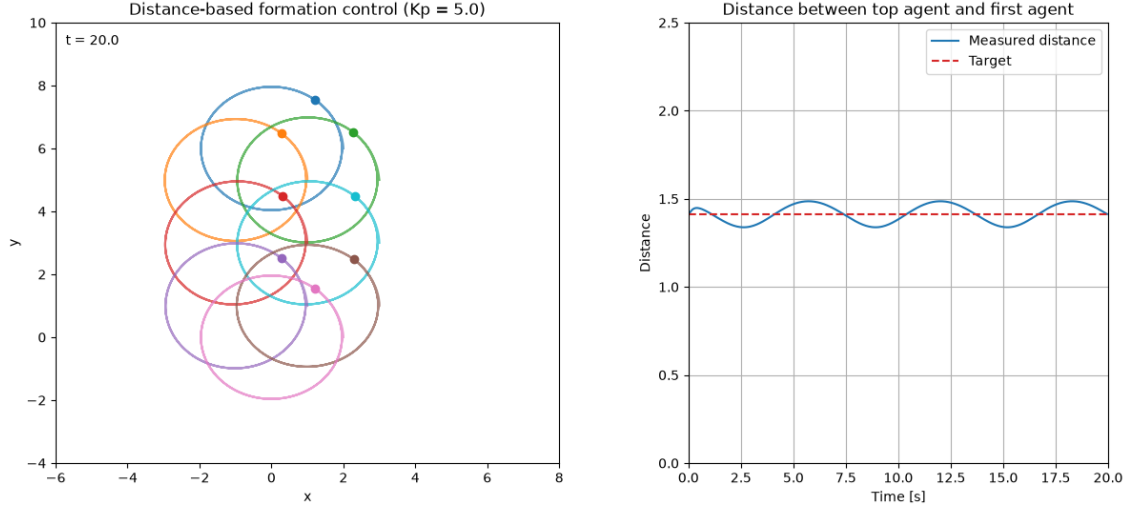


Figure 12: Agent paths with absolute position control of the top and bottom agents for $k_p = 5$.

Concluding Discussion

The exercises demonstrated several different approaches to formation control and the differences in behavior between them. In the position-based controller, the agents were able to follow their target trajectories while a constant wind disturbance produced a steady-state position error.

For the displacement-based controller, the desired formation was maintained while the motion of the leading agent propagated through the formation. This showed that the formation could be maintained even when each agent had access to only limited information, but it also made the propagation delays through the formation more apparent. Reversing the signs of the control laws also showed the importance of correct feedback since the system became unstable and diverged.

The distance-based controller showed that the formation can be controlled using only relative distances between neighboring agents. Here it became clear that distance constraints alone do not fully decide the orientation of the formation. At least two absolute-positioned anchors were needed to control the orientation of the formation.

AI Tools and AI Statement

Generative AI was used to help refactor the provided LaTeX template in accordance with the exercises. No report answers were generated by AI.

Generative AI was also used as a supplementary learning tool to improve the technical understanding of concepts covered in the exercises and to assist with minor code implementations.